

HYDROFARMS AGRI SOLUTIONS

GREENHOUSE COVERING,
POLYCARBONATE,
INSECT NETTING AND MOTORISED
SCREENS

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-ENV-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt - open terrain
Prepared by	Hydrofarms Agri Solutions
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Parent methods	HF-GH01-MST-STR-001 and HF-GH01-MST-CIV-002

WIND AND MOVING-SYSTEM CONTROL
No film, net or textile shall be pulled in unsuitable wind. External screens shall remain in the supplier-defined safe position whenever the wind/rain interlock requires it. No covering work may conceal an untested gutter, roof vent, fastener rail or structural interface.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, subcontractor control, programme and approval status.
Construction manager	Work-front release, access, sequencing and interface coordination.
Envelope specialist	Cutting plans, material orientation, tensioning, fixing and weather sealing.
Screen specialist	Zones, drives, shafts, supports, textile tracking and mechanical commissioning.
Structural engineer	Attachment loads, approved support points, roof-vent and gutter interfaces.
Electrical/controls engineer	Feeders, protection, isolators, sensors, commands and feedback.
QA/QC engineer	Batch traceability, inspection, tests, settings and turnover records.
HSE manager	Wind monitoring, work at height, lifting, sharp edges and emergency arrangements.
Client/consultant	Hold/witness inspection and final technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines receipt, storage, cutting, installation, tensioning, sealing, testing and handover of the greenhouse roof/side covering, service-greenhouse polycarbonate, insect-control net and complete motorised external/internal shading systems. It converts the approved product selections and zoning into a coordinated field procedure that preserves structural movement, drainage, ventilation, hygiene and maintenance access.

2. Scope and Package Boundary

This envelope package includes:

- SOLARWOVEN ULTRA to the production-greenhouse roof, three production sides and the service-greenhouse roof, including dedicated lock channels, spring locks, straps, tapes, seals and fixing fasteners.
- Clear/light-diffusing 10 mm multiwall polycarbonate to all four service-greenhouse sides, including H/U profiles, EPDM gaskets, closure tapes, compatible sealant and thermally tolerant fasteners.
- Xsect AirCool insect-control net to the approved service corridor/door barriers and external cooling-pad protection enclosure, including dedicated tensioning and closure accessories.
- HARMONY 5020 O E FR external motorised rolling screen and HARMONY 3015 O E internal motorised sliding screen, complete with mechanical drives, supports, limits, local isolators and commissioning.

PACKAGE BOUNDARY
Main steel, roof-vent structural frames, gutters and fan/pad chassis belong to structural packages. Upstream feeders, final power/control cables and central climate-system integration belong to electrical/automation. Groundcover belongs to the flooring/groundcover method. This method coordinates, but does not duplicate, those items.

3. Governing Information and Approval Priority

Priority	Controlled information
1	Applicable Egyptian statutory requirements and signed AFC structural/wind design
2	Approved envelope/screen shop drawings, cutting plans, fixing calculations and zone schedule
3	Latest approved manufacturer product data, installation instructions and warranties
4	This method statement, approved material samples and inspection/test plan
5	EN 13031-1 coordination basis for greenhouse structure/cladding attachment where adopted
6	EN 16153 or accepted equivalent for multiwall polycarbonate evidence
7	NEN 2675:2018 declared optical-property basis for selected screen products

NO IMPLIED CERTIFICATION

A referenced standard does not certify the supplied batch. Product identity, declared properties, warranty, FR claim where applicable, and compatibility of every accessory shall be evidenced by supplier submittals.

4. Project Geometry and Material Distribution

Element	Adopted basis	Installation consequence
Production greenhouse	126 m x 36 m; eaves 5.50 m; crown 7.00 m	Fourteen 9 m longitudinal bays
Service greenhouse	9 m x 36 m on west side	Roof film; four polycarbonate walls
Production plastic walls	Three sides only	Service interface is polycarbonate, not duplicated film
Exhaust openings	35 openings, nominal 1.40 x 1.40 m	Terminate and seal film to approved fan frames
Cooling-pad side	Approximately 125 m x 3 m opening	Terminate covering clear of pad frame and guttering
Roof vents	Consultant-approved operable vents	Complete movement/water tests before final closure
Internal screen elevation	Approximately 5.20 m AFFL	Maintain clearance to services and moving equipment

5. Approved Materials and Tender Quantities

Code	Material/system	Purchase basis	Key requirement
ENV-01	SOLARWOVEN ULTRA	6,670 m2	Reinforced greenhouse cover; orientation and seams by approved cutting plan
ENV-02	10 mm multiwall polycarbonate	45 sheets, 2.10 x 5.80 m	UV face outward; vertical cells; Polyee Plast or approved equal
ENV-03	Xsect AirCool net	850 m2 coordination allowance	Static insect barrier; no unsupported moving-door use
ENV-04	HARMONY 5020 O E FR external	5,105 m2	15 rolling zones; FR evidence required
ENV-05	HARMONY 3015 O E internal	5,105 m2	Five sliding zones; non-FR product

Quantities are procurement allowances. Final roll widths, seam positions, sheet cutting, net closures and fixing lengths shall be reconciled to approved shop drawings without changing the package boundary or duplicating another chapter.

5.1 Screen system properties

Property	External screen	Internal screen
Product	HARMONY 5020 O E FR	HARMONY 3015 O E
Material	76% polyolefin / 24% polyester	100% polyolefin
Weight	170 g/m2	110 g/m2
Strip width	4.6 mm	4 mm
Diffuse/direct PAR shading	61% / 58%	43% / 38%
Movement	Rolling, follows roof profile	Horizontal sliding at about 5.20 m
FR status	FR certificate required	Not FR classified

6. Screen Zoning and Mechanical Schedule

System	Approved zoning	Drive basis	Movement/storage
External	15 independent zones: 14 production bays + 1 service bay	15 x 0.75 kW geared motors	Across 36 m; roll storage at fan side
Internal	5 zones: production 4 + 3 + 3 + 4 bays, plus service	5 x 0.55 kW geared motors	Across 36 m; stacking at fan side

Component	Engineering quantity	Technical basis
External rolling shafts/tubes	135 m	HDG; 9 m per zone; final section by torque/deflection calculation
External bearing/support sets	105 sets	Sealed; supports at max. 1.50 m and shaft ends
External shaft couplings	30 sets	Keyed/clamped; full gearbox torque without slip
External support wires	10,260 m	UV/corrosion resistant; max. 0.50 m centres
External side guides/restraints	1,080 m	Anti-lift and anti-flutter for approved wind duty
Internal drive shafts	135 m	HDG keyed shafts; approved torsional-deflection check
Internal bearing/support sets	95 sets	Sealed; max. 1.50 m centres and zone ends
Internal shaft couplings	40 sets	Keyed/clamped corrosion-resistant type
Internal support wires	9,720 m	Max. 0.50 m centres; isolated from greenhouse steel
Internal synchronized drive	5 sets	Rack/pinion or stainless pull-wire system with tensioners
Wind / rain sensors	2 wind + 1 rain	Primary/standby wind logic and external safe-position command
Local isolation/E-stop	20 sets	One lockable IP65 station within safe sight of each motor

DRIVE APPROVAL

The complete screen package rate includes fabric, automatic movement mechanism, motors and gearboxes. Final shaft diameter, motor torque, gearbox ratio, support spacing and safe wind position require the screen specialist's signed calculations. Mechanical items shall not be priced again in electrical/automation.

7. Submittals Before Procurement

- Product data, product code, batch/roll identification, origin, warranty and physical sample for every textile, sheet, profile, gasket, tape and sealant.
- FR certificate specific to HARMONY 5020 O E FR; no FR claim shall be applied to SOLARWOVEN, Xsect AirCool or HARMONY 3015 O E unless separately certified.
- SOLARWOVEN roll widths, manufactured lengths, material orientation, seam/cutting plan, perimeter terminations and wind-fixing calculation.
- Polycarbonate light transmission, U-value, impact/fire data, UV-face identification, thermal expansion, compatible profiles/tapes/sealants and cutting layout.
- Net layout showing fixed barriers, pad enclosure, doors, overlaps, tensioning, cleaning access and closure of every structural/service penetration.

- Screen zone drawings showing each motor, shaft, coupling, bearing, support wire, guide, rack/cable, stacking area, sensor, isolator, limit, access and clearance.
- Motor and gearbox data, torque calculation, shaft deflection, connected load, wiring/control schematic, interlock narrative and spare-parts schedule.

8. Preconditions and Storage

- Steel alignment, bolts, coating repairs, covering rails and screen supports are accepted under HP-ENV-01; sharp projections and burrs are removed.
- Gutters, outlets, downpipes and roof vents have passed alignment, water and movement tests before the envelope conceals their interfaces.
- Fan/pad/door frames are complete, independently supported and dimensionally released; all perimeter terminations are accessible.
- Materials are stored dry, shaded and elevated. Rolls remain wrapped and supported; polycarbonate is flat, UV-face marked and protected from heat buildup.
- Manufacturer-approved weather window is confirmed. Wind speed is continuously monitored during lifting and pulling operations.
- MEWPs/scaffolds, soft slings, spreader bars, clean footwear, rescue equipment and protected cutting tables are available and inspected.

9. Detailed Installation Method

9.1 Setting out and interface release

- Transfer approved grid lines, zone boundaries, panel joints, roll seams, motor centres and stack/roll positions to the accepted structure.
- Check roof-vent envelopes, screen travel paths, gutters, fans, pipes, lighting and maintenance routes in the fully open and fully closed conditions.
- Install only at approved structural nodes. Drilling primary arches, local welding or attaching drives to covering rails without written structural approval is prohibited.

9.2 SOLARWOVEN ULTRA roof and sides

- Install compatible galvanized/aluminum lock channels, continuous protection tape at abrasion points and double fixing at perimeters, corners, doors, fans, pads and high-tension zones.
- Unroll on clean padded surfaces, orient by manufacturer markings and lift with soft, distributed support. Dragging over steel, concrete or fasteners is prohibited.
- Pull symmetrically in a suitable temperature/wind window. Set uniform tension without distortion, local hard spots, wrinkles, ponding pockets or restraint of structural movement.
- Secure with corrosion-resistant spring lock/wiggle wire and compatible channels. Fit hold-down/anti-flutter straps where required by the signed wind/fixing design.
- Form sealed, drained terminations around vents, gutters, fans, pads, doors and service penetrations. Use compatible UV-resistant repair tape only for accepted minor damage.

9.3 Service-greenhouse polycarbonate

- Confirm the external UV face and install sheets vertically with internal flutes vertical so condensation can drain downward.
- Cut on a supported clean bench; remove swarf from flutes without scratching. Seal upper flute ends with solid aluminum tape and lower ends with breathable anti-dust tape.
- Install continuous compatible EPDM gaskets and 10 mm aluminum H/U profiles. Provide manufacturer-required thermal clearance at joints, corners and fasteners.
- Use wide-washer corrosion-resistant fasteners with EPDM seals; predrill oversize movement holes where required and do not overtighten or crush cells.
- Apply neutral-cure polycarbonate-compatible sealant only. Acidic silicone, blocked flute drainage and unapproved solvent cleaners are prohibited.

9.4 Xsect AirCool insect barriers

- Install continuous fixed barriers at the service corridor, production/service separation and door interfaces up to the approved high point, plus the external pad enclosure shown on coordinated drawings.

- Tension evenly between dedicated rails/wires without reducing the designed ventilation free area. Reinforce corners, door jambs and service penetrations without open gaps.
- Use removable sealed collars or framed access panels at maintainable services. Close all holes immediately after work and inspect against daylight.
- Because the selected net is a static-use product, do not fix it to a moving door leaf without written manufacturer approval; use a fixed screened vestibule/barrier or approved moving-door mesh instead.

9.5 External motorised screen

- Install 15 aligned rolling zones, each corresponding to one 9 m longitudinal bay, with travel across the 36 m width and roll storage at the fan side.
- Set HDG rolling shafts, bearings and couplings to the approved centreline; support at maximum 1.50 m unless the signed supplier calculation requires closer spacing.
- Install support wires at maximum 0.50 m centres following the roof profile. Maintain at least 300 mm clear distance from roof covering through the full movement cycle.
- Fit side guides, end restraints and anti-lift/anti-flutter components. Provide self-locking 0.75 kW IP66 geared motors, dual limits, over-travel stop, overload/torque protection and manual override.

9.6 Internal motorised screen

- Install five horizontal sliding zones at approximately 5.20 m AFFL: production grouped 4 + 3 + 3 + 4 bays and one independent service-greenhouse zone.
- Set support wires at maximum 0.50 m centres and drive shafts/bearings to the approved geometry. Isolate textile and wires from sharp steel contact.
- Install synchronized rack-and-pinion or stainless pull-wire drives, leading profiles, take-up devices and stack restraints. Stacking remains at the fan side.
- Provide self-locking 0.55 kW IP55 geared motors, dual limits, over-travel stop, overload/torque protection, position feedback and manual override.

9.7 Local controls and safety interfaces

- Provide one local automatic package for 20 drives with percentage positioning, sequential starts, Auto-Off-Manual control, phase/overload protection and open/closed/fault feedback.
- Install one lockable IP65 isolator/emergency-stop station per motor within safe sight of its zone. Mechanical guards shall prevent access to shafts, couplings, racks and pinch points.
- Install two wind sensors in primary/standby arrangement and one rain sensor. Simulated high-wind/rain commands shall override normal operation and move the external screen to the supplier-approved safe position.
- Provide Modbus or dry-contact interface to the future central system; upstream power and final field cabling remain in electrical/automation BOQs.

10. Testing and Commissioning

- Inspect all coverings for product identity, orientation, uniform tension, undamaged surfaces, secure edges, sealed penetrations and free drainage.
- Water-test roof/gutter/vent and polycarbonate interfaces. Rectify leakage, trapped water, blocked flutes and uncontrolled discharge before acceptance.
- Inspect net barriers against daylight and smoke/airflow where approved; close every gap and verify doors/access panels return to a sealed condition.
- Record motor voltage/current, available torque, gearbox setting, shaft alignment, bearing temperature, limits, overload settings, manual operation and position feedback.
- Complete at least 20 automatic full open-close cycles for each of the 20 screen zones without jamming, abnormal noise, progressive misalignment, excessive sag, abrasion or textile damage.
- Demonstrate percentage positioning, synchronized travel, sequential starts, E-stops, local isolation, loss-of-phase/overload response and wind/rain safe-position logic.

11. Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	Product data, samples, warranties and certificates	Document/sample review	H	Material approval
02	Structure, gutters, vents and fixing rails released	Survey/visual/test records	H	Interface release
03	Delivered rolls, sheets, profiles and drives	Identity/batch/condition	W	Delivery inspection
04	Trial perimeter and representative fixing detail	Dimensional/visual	H	Mock-up approval
05	SOLARWOVEN installation and tension	Visual/dimensional	W	Cover report
06	Polycarbonate joints, tapes and movement clearance	Visual/water test	H	Cladding report
07	Insect barrier continuity	Visual/daylight	H	Biosecurity release
08	External shafts, wires, clearance and restraints	Survey/mechanical	H	External drive release
09	Internal shafts, wires and synchronized drive	Survey/mechanical	H	Internal drive release
10	Electrical protection, limits and local controls	Functional/electrical	W	Control checklist
11	Twenty-cycle test per zone and safety simulation	Performance test	H	Commissioning report
12	Final envelope, O&M, spares and as-builts	Record/final inspection	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

12. Hold Points

- HP-ENV-01: Signed shop drawings, samples, wind/fixing calculations and structural interfaces accepted before procurement/installation.
- HP-ENV-02: Gutters, roof vents, fan/pad/door frames and covering rails accepted before covering concealment.
- HP-ENV-03: Representative film perimeter/fixing mock-up accepted before general pulling.
- HP-ENV-04: Polycarbonate mock-up including H/U profiles, tapes, gasket, fastener and sealant accepted before repetition.
- HP-ENV-05: Net barrier and every penetration/door closure accepted before hygiene release.
- HP-ENV-06: External and internal mechanical drives accepted before motorized operation.
- HP-ENV-07: Twenty-cycle, safety-interlock and water tests accepted before handover.

13. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Wind / large textile sail	Weather window, anemometer, stop-work limit, controlled pull team and release plan	Medium
Work at height	MEWP/scaffold, fall restraint/arrest, rescue plan and exclusion zone	Medium
Dropped rolls/sheets	Lift plan, soft slings/spreader, tag lines and controlled access	Medium
Shafts and pinch points	Isolation/LOTO, guards, controlled jog mode and authorized specialist	Medium
Sharp profiles/sheets	Cut-resistant gloves, deburred edges, protected storage and handling	Low
Sealants/cleaners	Approved SDS, ventilation, gloves and compatible non-acidic products	Low
Waste/offcuts	Segregate plastic, PC, metal and packaging; secure light waste against wind	Low

14. Records, Spares and Handover

- Approved shop drawings, cutting/seam plans, zone schedules, structural attachment approval and supplier calculations.
- Product certificates, FR evidence, batch/roll records, warranties, samples and delivery inspections.
- Cover, polycarbonate, net, water-test and closure inspection reports with repair register.
- Motor/gearbox/shaft/bearing settings, electrical checks, sensor calibration and 20-cycle reports by zone code.
- As-built drawings showing seams, panels, net barriers, motors, shafts, sensors, isolators, limits and safe position.
- O&M manuals, cleaning restrictions, repair tools, compatible tapes/sealants, spare textile/sheets, bearings, limits and recommended critical spares.
- Operator training covering normal positioning, weather override, emergency stop, manual recovery, inspection and safe isolation.